

INNOVA JUNIOR COLLEGE  
JC 2 PRELIMINARY EXAMINATION  
in preparation for General Certificate of Education Advanced Level  
**Higher 2**

CANDIDATE  
NAME

CLASS

INDEX NUMBER

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**CHEMISTRY**

**9729/01**

Paper 1 Multiple Choice

**14 Sep 2018**

**1 hour**

Additional Materials:

*Data Booklet*

Multiple Choice Answer Sheet

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**READ THESE INSTRUCTIONS FIRST**

Write your index number, name and class on all the work you hand in.

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

**Read the instructions on the Answer Sheet very carefully.**

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

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This document consists of **16** printed pages and **1** blank page.



Innova Junior College

**[Turn over**

For each question there are four possible answers, **A**, **B**, **C**, and **D**. Choose the **one** you consider to be correct.

- 1 In an experiment, 0.002 mol acidified iron (II) sulfate is quantitatively oxidised by 0.001 mol of an oxidising agent,  $\text{ZO}_3^-$ . Given that  $\text{Fe}^{2+}$  is oxidised to  $\text{Fe}^{3+}$ , what is the final oxidation state of **Z**?

**A** +1                      **B** +2                      **C** +3                      **D** +4

- 2 Gaseous particle **X** has a proton number  $n$  and a charge of +1.

Gaseous particle **Y** has a proton number of  $(n + 1)$  and is isoelectronic with **X**.

Which of the following statements correctly describe **X** and **Y**?

- 1 **Y** has a smaller radius than **X**.
- 2 **Y** requires less energy than **X** when a further electron is removed from each particle.
- 3 **Y** releases less energy than **X** when an electron is added to each particle.

**A** 1 only  
**B** 2 only  
**C** 1 and 2 only  
**D** 2 and 3 only

- 3 What is the electronic configuration of an element with a **second** ionisation energy higher than that of each of the elements either side of it in the Periodic Table?

**A**  $1s^2 2s^2 2p^6 3s^2$   
**B**  $1s^2 2s^2 2p^6 3s^2 3p^1$   
**C**  $1s^2 2s^2 2p^6 3s^2 3p^2$   
**D**  $1s^2 2s^2 2p^6 3s^2 3p^3$

- 4 The table shows the charge and radius of each of six ions.

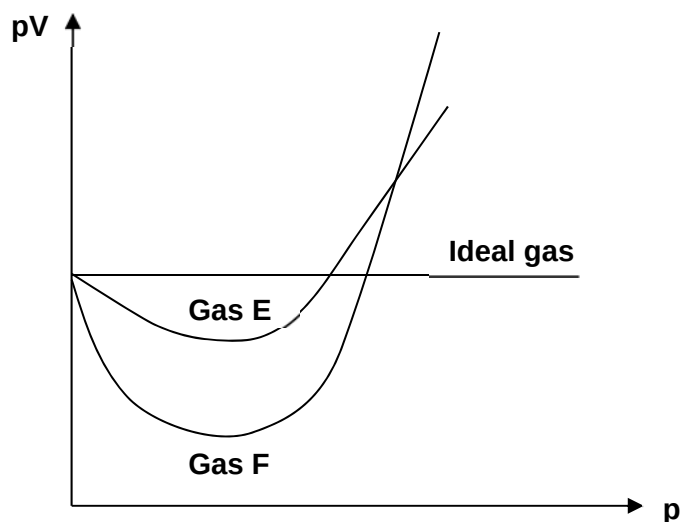
| ion         | $J^+$ | $L^+$ | $M^{2+}$ | $X^-$ | $Y^-$ | $Z^{2-}$ |
|-------------|-------|-------|----------|-------|-------|----------|
| radius / nm | 0.14  | 0.18  | 0.15     | 0.14  | 0.18  | 0.15     |

The ionic solids JX, LY and MZ are of the same lattice type. What is the correct order of their lattice energies, placing the most exothermic first?

- A** JX, MZ, LY  
**B** LY, MZ, JX  
**C** MZ, JX, LY  
**D** MZ, LY, JX
- 5 In certain microwave ovens, the wave energy produced is absorbed by certain polar molecules.

Which of the following would absorb microwave energy?

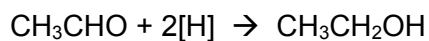
- 1  $CH_3CH_2OH$   
 2  $AlCl_3$   
 3  $CO_2$   
 4  $CH_3F$
- A** 1 and 2 only  
**B** 1 and 4 only  
**C** 2 and 3 only  
**D** 3 and 4 only



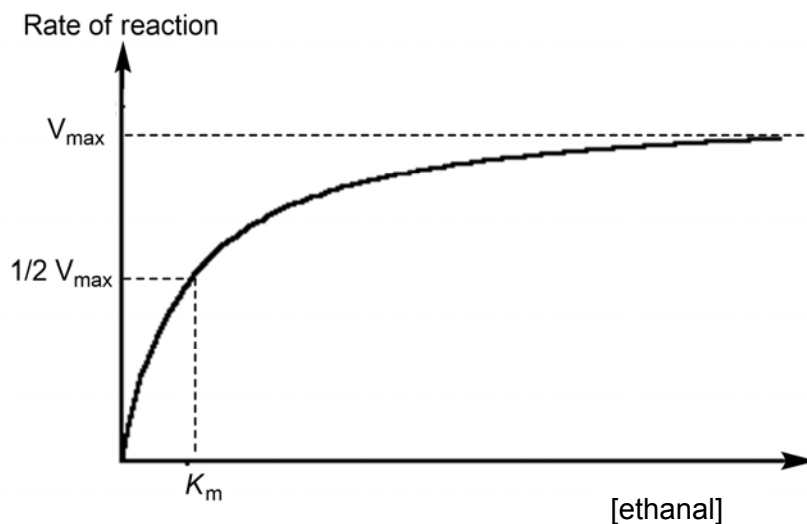
Which one of the following statements is true for the observation?

- A Gas E could be He while gas F could be  $\text{CO}_2$ .
- B Gas E could be  $\text{NH}_3$  while gas F could be  $\text{H}_2$ .
- C Gas E could form intermolecular hydrogen bonds while gas F could form intermolecular permanent dipole-permanent dipole.
- D Gas E could form stronger intermolecular instantaneous dipole-induced dipole than gas F.

- 7 The enzyme alcohol dehydrogenase catalyses an important step in the production of ethanol by fermentation.



The graph shows how the rate of this enzyme-catalysed reaction varies with the concentration of ethanal.

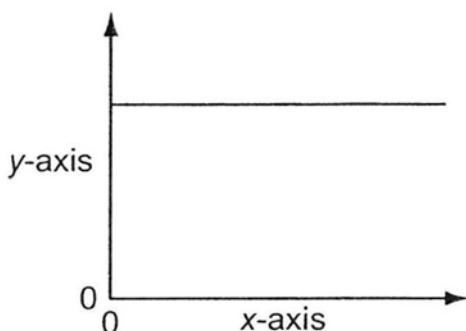


Which statement best explains the reason for the flattening off of the curve?

- A All the ethanal has been used up and the reaction has finished.
- B As the ethanol product builds up the reaction slows down.
- C At high ethanal concentrations all the active sites in the enzyme molecules are occupied by ethanal molecules.
- D At high ethanal concentrations the ethanal inhibits the action of enzyme.

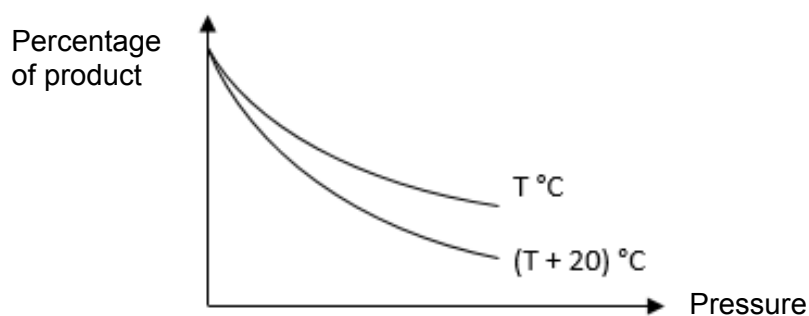
- 8 The kinetics of the zero order reaction  $P \rightarrow Q$  were investigated under different conditions. The table shows pairs of quantities that were plotted as graphs.

Which pairs gave the following graph?



|   | y-axis        | x-axis        |
|---|---------------|---------------|
| 1 | rate          | time          |
| 2 | rate constant | time          |
| 3 | rate constant | temperature   |
| 4 | rate          | concentration |

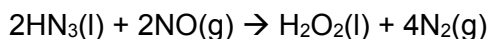
- A 1 and 4 only  
 B 2 and 3 only  
 C 1, 2 and 3 only  
 D 1, 2 and 4 only
- 9 The graph below shows how the percentage of product present at equilibrium varies with temperature and pressure for a reaction.



Which reaction could the graph represent?

- A  $4\text{Fe(s)} + 3\text{O}_2\text{(g)} \rightleftharpoons 2\text{Fe}_2\text{O}_3\text{(s)}$   $\Delta H = -1644 \text{ kJ mol}^{-1}$   
 B  $2\text{C(s)} + \text{O}_2\text{(g)} \rightleftharpoons 2\text{CO(g)}$   $\Delta H = -222 \text{ kJ mol}^{-1}$   
 C  $\text{N}_2\text{O}_4\text{(g)} \rightleftharpoons 2\text{NO}_2\text{(g)}$   $\Delta H = +57.2 \text{ kJ mol}^{-1}$   
 D  $\text{CO(g)} + \text{Cl}_2\text{(g)} \rightleftharpoons \text{COCl}_2\text{(s)}$   $\Delta H = +86 \text{ kJ mol}^{-1}$

- 10 The standard enthalpy change of the following reaction is  $-896.4 \text{ kJ mol}^{-1}$ .

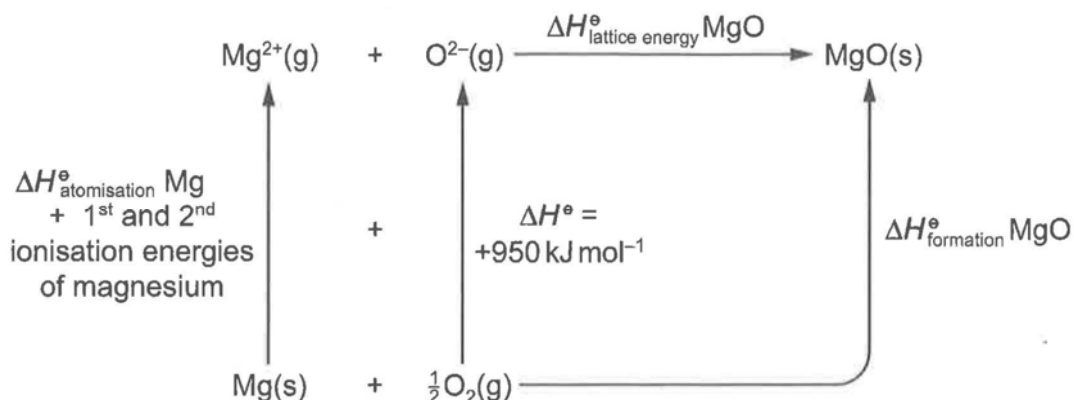


|                                          | NO(g)  | H <sub>2</sub> O <sub>2</sub> (l) |
|------------------------------------------|--------|-----------------------------------|
| $\Delta H_f^\theta / \text{kJ mol}^{-1}$ | + 90.3 | - 187.8                           |

Using the standard enthalpy change of formation in the above table, what is the standard enthalpy change of formation, in  $\text{kJ mol}^{-1}$ , of  $\text{HN}_3(\text{l})$ ?

- A +264  
 B +528  
 C +618  
 D +632
- 11 A student calculated the lattice energy for magnesium oxide using the cycle shown.

The enthalpy change of atomisation of magnesium,  $\Delta H_{\text{atomisation}}^\theta \text{Mg}$ , is the energy needed when 1 mol of gaseous magnesium atoms is formed from 1 mol of solid magnesium under standard conditions.



However, the value calculated by the student for the lattice energy was **less** exothermic than the correct value.

Which errors could have been made in the calculation?

- 1 omitting the enthalpy change of atomisation of magnesium
  - 2 omitting the first ionisation energy of magnesium
  - 3 using the standard enthalpy change of combustion of magnesium rather than the standard enthalpy change of formation of magnesium oxide
- A 1 and 2 only  
 B 1 and 3 only  
 C 2 and 3 only  
 D 1, 2 and 3

12 Which changes are accompanied by an increase in entropy of the system?

- 1 freezing of water
- 2 sublimation of iodine
- 3 vaporisation of ethanol
- 4 conversion of  $\text{O}_2(\text{g})$  to  $\text{O}_3(\text{g})$

- A 1 and 2 only
- B 2 and 3 only
- C 3 and 4 only
- D 1 and 4 only

13 The value of the ionic product of water,  $K_w$ , varies with temperature.

| Temperature / °C | $K_w / \text{mol}^2 \text{dm}^{-6}$ |
|------------------|-------------------------------------|
| 25               | $1.0 \times 10^{-14}$               |
| 62               | $1.0 \times 10^{-13}$               |

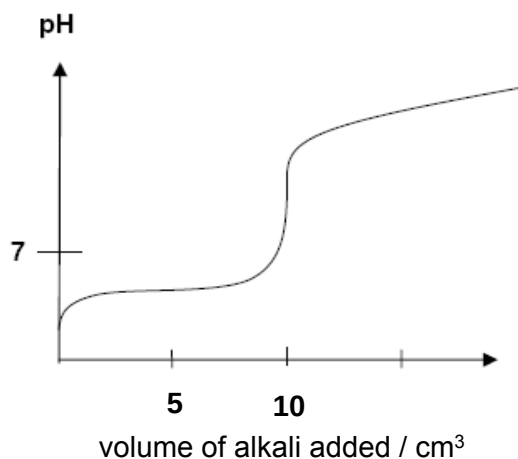
What can be deduced from this information?

- A The ionic dissociation of water is an exothermic process.
- B The association of water molecules by hydrogen bonding increases as temperature increases.
- C The pH of pure water increases with temperature.
- D At 62 °C, water with a pH of 6.5 is neutral.



- 14 The following graph shows the changes in pH of 20.0 cm<sup>3</sup> of 1.0 mol dm<sup>-3</sup> acid solution when excess 1.0 mol dm<sup>-3</sup> alkali solution is added gradually.

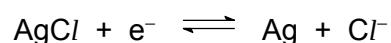
Which one of the following pairs of solution with a suitable indicator could have resulted in the graph below?



|   | alkali              | acid                                              | indicator       |
|---|---------------------|---------------------------------------------------|-----------------|
| A | NaOH                | H <sub>2</sub> SO <sub>4</sub>                    | Methyl orange   |
| B | NH <sub>3</sub>     | HNO <sub>3</sub>                                  | Phenolphthalein |
| C | KOH                 | CH <sub>3</sub> CO <sub>2</sub> H                 | Methyl orange   |
| D | Ba(OH) <sub>2</sub> | CH <sub>3</sub> CH <sub>2</sub> CO <sub>2</sub> H | Phenolphthalein |

- 15 Use of the Data Booklet is relevant to this question.

The AgCl / Ag reference electrode is based on the following reaction:



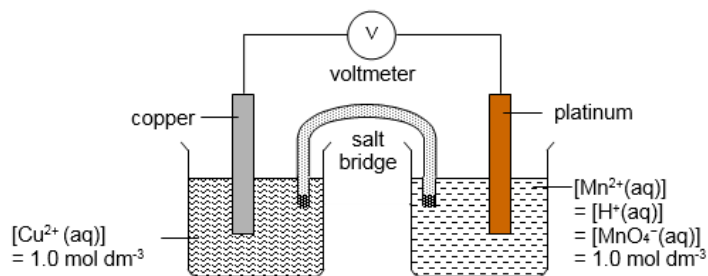
The reduction potential of the Fe<sup>2+</sup> / Fe half-cell was found to be -0.64 V when measured using the AgCl / Ag reference electrode.

What is the reduction potential of the AgCl / Ag electrode with respect to the standard hydrogen electrode?

- A    -1.10 V                      B    -0.20 V                      C    +0.20 V                      D    +1.10 V

- 16 Use of the Data Booklet is relevant to this question.

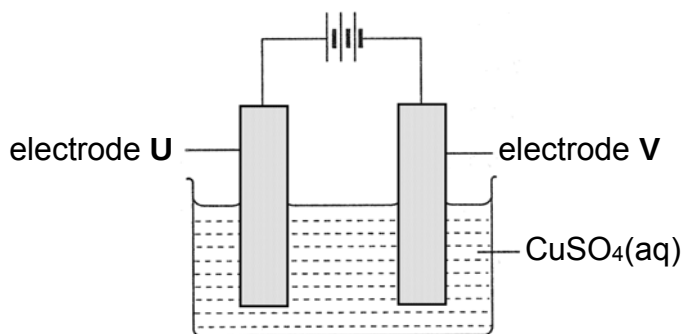
A cell is set up by connecting a  $\text{Cu}^{2+}/\text{Cu}$  half-cell and an acidified  $\text{MnO}_4^-/\text{Mn}^{2+}$  half-cell.



Which change in the set up will cause the corresponding effect on the e.m.f (potential difference) of the cell?

|          | change                                                                    | effect on e.m.f of cell |
|----------|---------------------------------------------------------------------------|-------------------------|
| <b>A</b> | addition of excess $\text{NH}_3$ (aq) into oxidation half-cell            | increases               |
| <b>B</b> | addition of concentrated $\text{H}_2\text{SO}_4$ into reduction half-cell | decreases               |
| <b>C</b> | increase the size of the copper electrode                                 | increases               |
| <b>D</b> | replace copper with an alloy of copper and zinc                           | remains the same        |

- 17 In an experiment, a cell was set up to obtain pure copper from a copper-silver alloy as shown below. Electrode **V** is pure copper and electrode **U** is the copper-silver alloy.

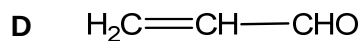
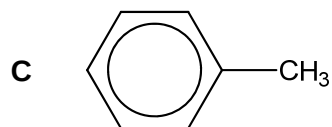
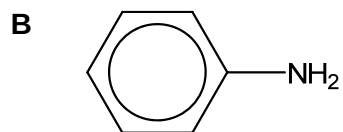
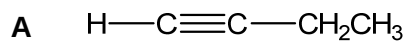


When a current of 40.0 A flows through the electrolyte for 26.8 minutes, the mass of the anode changes by 26.47 g.

What is the percentage of silver by mass in the electrode **U**?

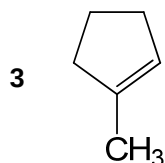
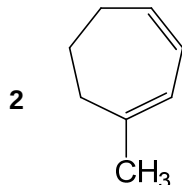
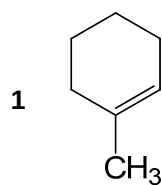
- A** 20%
- B** 40%
- C** 60%
- D** 80%

18 Which of the following is a planar molecule?



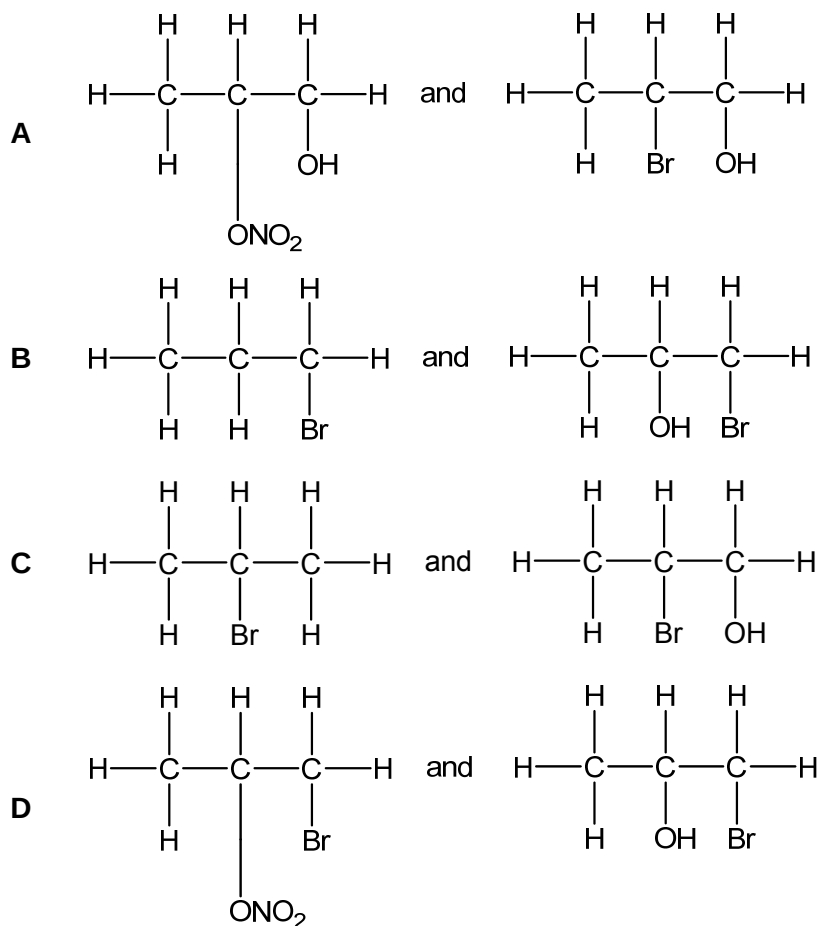
19 A hydrocarbon, on heating with an excess of hot concentrated acidic  $\text{KMnO}_4$  produces  $\text{HO}_2\text{CCH}_2\text{CH}_2\text{CH}_2\text{COCH}_3$ .

What could the hydrocarbon be?



- A 1 and 2 only  
 B 1 and 3 only  
 C 2 and 3 only  
 D 1, 2 and 3

- 20 When propene reacts with  $\text{Br}_2$  in the presence of excess aqueous  $\text{KNO}_3$ , what are the two major products formed?



- 21 The molecule of benzene,  $\text{C}_6\text{H}_6$  is a regular hexagon in which the  $\pi$  electrons are described as delocalised.

Which of the following statements are true for benzene?

- 1 It prevents benzene from undergoing addition reactions.
  - 2 All C–C bonds lengths in benzene are intermediate between C–C bond in an alkane and C=C bond in an alkene.
  - 3 The enthalpy change of hydrogenation of benzene is less exothermic than that predicted for cyclohexatriene.
  - 4 The presence of  $\pi$  electrons enables benzene to be a good electrical conductor.
- A 1 and 2 only
- B 2 and 3 only
- C 1, 2 and 3 only
- D 1, 2 and 4 only

- 22 Some chlorobutanes were separately treated with hot ethanolic sodium hydroxide. Two of these gave the same hydrocarbon,  $C_4H_6$ .

From which pair of chlorobutanes was this hydrocarbon obtained?

- A  $CH_3CH_2CH_2CH_2Cl$  and  $CH_3CH_2CH_2CHCl_2$   
 B  $CH_3CH_2CH_2CH_2Cl$  and  $ClCH_2CH_2CH_2CH_2Cl$   
 C  $CH_3CH_2CCl_2CH_3$  and  $ClCH_2CH_2CH_2CHCl_2$   
 D  $CH_3CHClCHClCH_3$  and  $ClCH_2CH_2CH_2CH_2Cl$

- 23 Deuterium, D, is the  $^2_1H$  isotope of hydrogen.

Which of the following reactions yield a carbon compound containing deuterium?

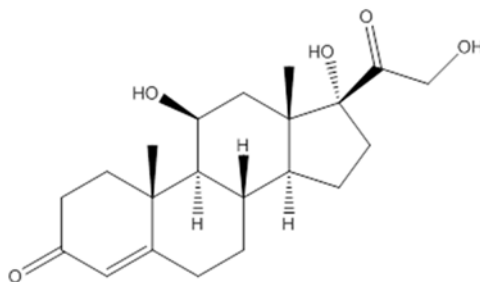
- 1  $CH_3CH_2CN \xrightarrow[\text{heat}]{D_2, \text{Ni catalyst}}$   
 2  $CH_3CO_2CH_2CH_3 \xrightarrow[\text{heat}]{D_2SO_4, D_2O}$   
 3  $CH_3COCl \xrightarrow[D_2O]{NaOD}$   
 4  $(CH_3)_3COH \xrightarrow[\text{heat}]{\text{conc } D_2SO_4}$

- A 1 and 2 only  
 B 1 and 4 only  
 C 1, 2 and 3 only  
 D 2, 3 and 4 only

- 24 Which of the following synthetic routes **does not** produce ethanoic acid?

- A  $CH_3Br \xrightarrow[\text{heat}]{\text{alcoholic KCN}} \xrightarrow[\text{heat}]{\text{dil HCl}}$   
 B  $CH_3CHO \xrightarrow[\text{warm}]{[Ag(NH_3)_2]^+} \xrightarrow[\text{r.t.p.}]{H^+}$   
 C  $CH_3CH(OH)CH_3 \xrightarrow[\text{warm}]{\text{aqueous alkaline iodine}} \xrightarrow[\text{r.t.p.}]{H^+}$   
 D  $CH_3CH(OH)CH_2CH_3 \xrightarrow[180^\circ C]{\text{excess conc } H_2SO_4} \xrightarrow[\text{heat}]{\text{acidic } K_2Cr_2O_7}$

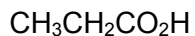
- 25 Hydrocortisone is a steroid hormone produced by the adrenal gland and is released in response to stress. It is commonly used as an active ingredient in anti-inflammatory creams.



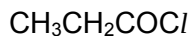
Hydrocortisone

Which of the following statements about Hydrocortisone is true?

- A** When treated with  $\text{NaBH}_4$  in the presence of methanol, it forms a compound containing 5 hydroxy groups.
- B** When warmed with aqueous alkaline iodine, a yellow precipitate is observed.
- C** When treated with cold dilute  $\text{KMnO}_4$ , it forms a compound containing 2 hydroxy groups.
- D** When treated with an excess of hot concentrated acidified  $\text{KMnO}_4$ , it forms a compound containing 7 chiral centres.
- 26 Equal amounts of compounds **P**, **Q**, **R** and **S** are separately shaken with  $100 \text{ cm}^3$  of water. The pH of each resultant solution is then measured.



**P**



**Q**



**R**

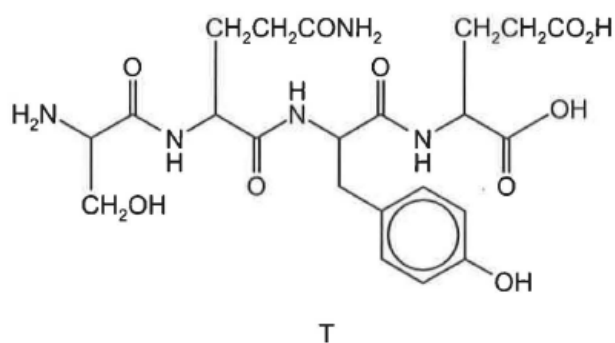


**S**

Which of the following shows the correct order of increasing order of acidity of the solutions formed?

- A** P, R, S, Q
- B** P, S, R, Q
- C** Q, P, S, R
- D** Q, S, R, P

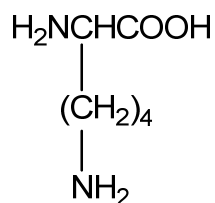
- 27** The diagram shows the structure of the tetrapeptide T.



When 0.1 mole of **T** is heated under reflux with NaOH(aq) until no further reaction occurs, how many moles of NaOH will react?

- |          |     |          |     |
|----------|-----|----------|-----|
| <b>A</b> | 0.4 | <b>B</b> | 0.5 |
| <b>C</b> | 0.6 | <b>D</b> | 0.7 |

- 28** Lysine is an  $\alpha$ -amino acid.



Which structure is predominant when lysine is in an aqueous solution of pH 9.5, given that lysine has three  $pK_a$  values of 2.2, 8.9 and 10.5?

- |          |                                                                                                                        |          |                                                                                                         |
|----------|------------------------------------------------------------------------------------------------------------------------|----------|---------------------------------------------------------------------------------------------------------|
| <b>A</b> | $\begin{array}{c} \text{H}_3\text{N}^+\text{CHCO}_2\text{H} \\   \\ (\text{CH}_2)_4 \\   \\ \text{NH}_3^+ \end{array}$ | <b>B</b> | $\begin{array}{c} \text{H}_2\text{NCHCO}_2^- \\   \\ (\text{CH}_2)_4 \\   \\ \text{NH}_2 \end{array}$   |
| <b>C</b> | $\begin{array}{c} \text{H}_3\text{N}^+\text{CHCO}_2^- \\   \\ (\text{CH}_2)_4 \\   \\ \text{NH}_3^+ \end{array}$       | <b>D</b> | $\begin{array}{c} \text{H}_2\text{NCHCO}_2^- \\   \\ (\text{CH}_2)_4 \\   \\ \text{NH}_3^+ \end{array}$ |

**29** Element **J** is in Period 3 of the Periodic Table. The following three statements are the properties of element **J** or its compounds.

- Adding NaOH(aq) to the solution resulting from the reaction of a chloride of **J** with water produces a white precipitate which is soluble in an excess of NaOH(aq).
- Element **J** is a solid at room temperature.
- The oxide of element **J** is soluble in hydrochloric acid.

What is the identity of element **J**?

- |                    |                     |
|--------------------|---------------------|
| <b>A</b> magnesium | <b>C</b> silicon    |
| <b>B</b> aluminium | <b>D</b> phosphorus |

**30** *The use of the Data Booklet is relevant to this question.*

The colours of various vanadium ions in aqueous solution are given in the table below.

| Oxidation state | V               | IV               | III             | II              |
|-----------------|-----------------|------------------|-----------------|-----------------|
| Ion             | $\text{VO}_2^+$ | $\text{VO}^{2+}$ | $\text{V}^{3+}$ | $\text{V}^{2+}$ |
| Colour          | yellow          | blue             | green           | violet          |

What is the final colour of the solution when excess zinc powder is added to an acidified solution containing  $\text{VO}_2^+$  ions?

- |                 |                 |
|-----------------|-----------------|
| <b>A</b> yellow | <b>C</b> green  |
| <b>B</b> blue   | <b>D</b> violet |

**End of Paper**



